Views - ***Assignment 11***

1) Create a view that shows all of the customers who have the highest ratings.

```
d6_96728_shaurya> SELECT * FROM customers WHERE rating =(select rating from
customers ORDER BY RATING DESC LIMIT 1);
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
2 rows in set (0.006 sec)

d6_96728_shaurya>
```

```
d6_96728_shaurya> SELECT * FROM v1_emp;
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
2 rows in set (0.007 sec)

d6_96728_shaurya>
```

2) Create a view that shows the number of salespeople in each city.

```
d6_96728_shaurya> select city , count(snum) AS People_in_City from salespeople group by (city);
+-----------+----------------+
| city      | People_in_City |
+-----------+----------------+
| London    |              2 |
| San Jose  |              1 |
| Barcelona |              1 |
| New York  |              1 |
+-----------+----------------+
4 rows in set (0.006 sec)

d6_96728_shaurya> create view People_in_City AS
    -> select city , count(snum) AS People_in_City from salespeople group by (city);
Query OK, 0 rows affected (0.081 sec)

d6_96728_shaurya> select * from People_in_City;
+-----------+----------------+
| city      | People_in_City |
+-----------+----------------+
| London    |              2 |
| San Jose  |              1 |
| Barcelona |              1 |
| New York  |              1 |
+-----------+----------------+
4 rows in set (0.026 sec)

d6_96728_shaurya>
```

3) Create a view that shows the average and total orders for each salesperson after his or her name. Assume all names are unique.

STEP 1 :

```
d6_96728_shaurya> select count(onum), snum from orders GROUP BY snum ORDER BY SNUM ASC;
+-------------+------+
| count(onum) | snum |
+-------------+------+
|           3 | 1001 |
|           3 | 1002 |
|           1 | 1003 |
|           1 | 1004 |
|           2 | 1007 |
+-------------+------+
```

STEP 2 :

```
d6_96728_shaurya> select * from salespeople s left join (select count(onum), snum from orders GROUP BY snum ORDER BY SNUM ASC) as sa on sa.snum
= s.snum;
+------+---------+-----------+------+------------+------+
| snum | sname   | city      | comm | count(onum) | snum |
+------+---------+-----------+------+------------+------+
| 1001 | Peel    | London    | 0.12 |          3 | 1001 |
| 1002 | Serres  | San Jose  | 0.13 |          3 | 1002 |
| 1004 | Motika  | London    | 0.11 |          1 | 1004 |
| 1007 | Rifkin  | Barcelona | 0.15 |          2 | 1007 |
| 1003 | Axelrod | New York  | 0.10 |          1 | 1003 |
+------+---------+-----------+------+------------+------+
5 rows in set (0.008 sec)
```

STEP 3:

```
d6_96728_shaurya> select snum, count(onum) as total_orders, AVG(AMT) A
S average_sale from ORDERS GROUP BY snum ORDER BY SNUM;
+------+--------------+--------------+
| snum | total_orders | average_sale |
+------+--------------+--------------+
| 1001 |            3 |  5127.356667 |
| 1002 |            3 |  1848.716667 |
| 1003 |            1 |  1713.230000 |
| 1004 |            1 |  1900.100000 |
| 1007 |            2 |   558.425000 |
+------+--------------+--------------+
5 rows in set (0.006 sec)
```

STEP 4:

```
d6_96728_shaurya> select sname, total_orders, average_sale  from sales
people s left join (select snum, count(onum) as total_orders, AVG(AMT)
 AS average_sale from ORDERS GROUP BY snum ORDER BY SNUM) as sa on s.s
num = sa.snum;
+---------+--------------+--------------+
| sname   | total_orders | average_sale |
+---------+--------------+--------------+
| Peel    |            3 |  5127.356667 |
| Serres  |            3 |  1848.716667 |
| Motika  |            1 |  1900.100000 |
| Rifkin  |            2 |   558.425000 |
| Axelrod |            1 |  1713.230000 |
+---------+--------------+--------------+
5 rows in set (0.008 sec)
```

STEP 5:

```
d6_96728_shaurya> SELECT * FROM AVG_SALES;
+---------+--------------+--------------+
| sname   | total_orders | average_sale |
+---------+--------------+--------------+
| Peel    |            3 |  5127.356667 |
| Serres  |            3 |  1848.716667 |
| Motika  |            1 |  1900.100000 |
| Rifkin  |            2 |   558.425000 |
| Axelrod |            1 |  1713.230000 |
+---------+--------------+--------------+
```

4) Create a view that shows each salesperson with multiple customers.

STEP 1:

```
d6_96728_shaurya> select * from vss;
+---------+----------+
| sname   | cname    |
+---------+----------+
| Peel    | Clemens  |
| Peel    | Hoffman  |
| Serres  | Grass    |
| Serres  | Liu      |
| Motika  | Pereira  |
| Rifkin  | Cisneros |
| Axelrod | Giovanni |
+---------+----------+
7 rows in set (0.008 sec)

d6_96728_shaurya>
```

Changing Values through Views.

5) Which of these views are updateable (will allow DML operations)?

#1 Create View Dailyorders
as Select Distinct cnum, snum, onum, odate from Orders;
Not Updateable
Reason: Uses **DISTINCT,** Views with DISTINCT are **not updateable**

#2 Create View Custotals
as Select cname, Sum (amt) Sum_Amt from Orders, Customers
where Orders.cnum=Customers.cnum Group by cname;
Reasons: Uses **SUM() aggregate,** Uses **GROUP BY,** Uses **join of two tables**
Any of these alone already makes the view **non-updateable**.

#3 Create view Thirdorders

as Select * from Dailyorders where odate='1990-10-03';

Not Updateable

Reason : Based on another **non-updateable view (Dailyorders)**.

#4 Create view Nullcities
as Select snum, sname, city
from Salespeople
where city is NULL
OR sname BETWEEN 'A' and 'MZ';

Updateable Reason:

Single table
No aggregate
No GROUP BY
No DISTINCT So **DML operations allowed**.

6) Create a view of the Salespeople table called Commissions. This view will include only the snum and comm fields. Through this view, someone could enter or change commissions, but only to values between .10 and .20.

7)

```
d6_96728_shaurya> CREATE VIEW Commissions AS
    -> SELECT snum, comm
    -> FROM Salespeople
    -> WHERE comm BETWEEN 0.10 AND 0.20
    -> WITH CHECK OPTION;
Query OK, 0 rows affected (0.141 sec)

d6_96728_shaurya> select * from Commissions;
+------+------+
| snum | comm |
+------+------+
| 1001 | 0.12 |
| 1002 | 0.13 |
| 1004 | 0.11 |
| 1007 | 0.15 |
| 1003 | 0.10 |
+------+------+
5 rows in set (0.127 sec)
```